My thermal stochastic risk engine simulates how random variables could affect thermal sensor readings and stock prices using a Monte Carlo approach. This project blends engineering and finance.

In the first part of my project, I simulate how noise interference can affect temperature reading instruments when a CPU goes from starting temperature to operating temperature. I used Newton's Law of Cooling to achieve this. I also included a max safe temperature limit to see if the CPU ever crosses the limit during simulation.

In the second part of my project, I simulated how stock prices can fluctuate using geometric Brownian motion. I used the stochastic differential equation for geometric Brownian motion after using Ito calculus equation to achieve this. In this section, I also reuse the quant engine to run a simulation on the chances of thermal failure of the CPU heating up and crossing the safe limit here.

In the third part of my project, I create visualizations of the data produced by the models and plot them on a graph. The first part of the project will create a graph of the heating data when the CPU is moving towards operating temperature and the second part of the project will create a graph that shows all the paths a stock could take with the geometric Brownian motion and highlighting the mean price. I also added an additional part that calculates the value at risk and the conditional value at risk.

Finally, I created a main project file to run all the parts of the project.

Thermal_model

For my thermal cooling loop, I chose to have ambient temp be typical room temperature which is around 20°C and according to ThermalStats, critical temperature for a CPU is around 90°C.

I chose the number of steps to be 252 as there are 252 trading days in a year.

A CPU’s normal temperature is around 40°C, so that will be my initial temperature, and around 80°C is a warm GPU when gaming (ThermalStats Team, 2026). This simulation will simulate a computer going from normal tasks to gaming immediately and seeing its effects.

Thermocouples have an uncertainty of ±0.5°C to ±2°C, so I chose a noise STD of 1.25 (2+0.5)/2 (Doebelin, 2004).

This phase of the project is based on Newton’s law of cooling:

$$T(t) = T_{env} + (T(0) - T_{env})e^{-\frac{t}{\tau}}$$

and since $k = \frac{1}{\tau}$, we can write it as

$$T(t) = T_{env} + (T(0) - T_{env})e^{-kt}$$

$T(t)$ is the

T_{env}
T_0 is the initial temperature

k Is the heating coefficient

I assumed on average that it would take around 20 minutes to get a CPU from room temperature up to our target temperature, so I calculated the cooling constant from the formula which equals 0.05.

Finally, I simulated the sensor noise and then added that to our calculated theory temperature to get our final temperature.

quant_engine

For the number of simulations, I just chose 1000. 1000 feels big enough. I chose 252 time steps to match the thermal model.

For this section I will be using the stochastic differential equation for geometric Brownian motion after using Ito calculus:

$$S(t) = S(0)^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B(t)}$$

$$B(t) \sim \sqrt{(dt)}N(0,1)$$

$N(0,1)$ is a normal distribution with zero mean and unit variance so we would have a $\mu = 0$ and $\sigma = 1$ and we will call that Z.

I chose 100 to be my initial stock price ($S(0)$) so it would be easy to see how much the stock price would go up by %. The average return of the S&P 500 after adjusting for inflation is 7.2%, so I just rounded to 7% and made that my expected annual return () (Min, 2026). As of 8/8/26, the S&P historical volatility () is 12.89% so I rounded it up to 13% (*S&P 500 Index Volatility History & Chart Since 1974*, 2026).

In the visualizer, I'm also adding a section to calculate the value at risk and the conditional value at risk. This would be useful to see what the chances the portfolio underperforms and how much it could lose if it does. I'm using this value at risk formula from Ryan O'Connell's (2026) website:

$$VaR_{\alpha} = -Portfolio\,Value \cdot Percentile_{(1-\alpha)}(R)$$

A good confidence is 95% so my confidence level will be set at 0.95 (Harper, 2025).

Conditional value at risk is:

$$ES_{\alpha} = E\,[L\,|\,L\,\geq\,VaR_{\alpha}|]$$

This basically just means out of the returns, how much did it cross the bad percentile.

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